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Managing agricultural water use in the Western United States: A search for efficient climate adaptation institutions

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ABSTRACT

Irrigated agriculture throughout much of the western United States relies on water resources that are under increasing stress from more frequent and severe droughts, reduced snowpack, and changes in the timing of runoff. Much of the water used for irrigation in that region passes through lands managed by irrigation organizations. These organizations play an important role in the use, allocation, and conservation of water resources through the maintenance of conveyance infrastructure, administration of rights to use water, and distribution of water to users. However, little work has been published to date systematically describing economic effects from alternative irrigation water trading arrangements for adapting to drought and climate change. The contribution of this work is to fill a gap in previous research by assessing the economic performance of various water trading arrangements for adapting to water shortages. We secured U.S. Department of Agriculture data on irrigation water use patterns in the central part of the Rio Grande Basin of the American Southwest to support formulation of a mathematical model predicting water use patterns to assess impacts on farm income associated with a range of water shortages and water trading arrangements for adapting to these shortages. Model results show that unlimited water trading minimizes farm income lost from shortages, while a more limited geographic scope for trading also moderates farm income losses. Among the three shortage sharing methods considered, a proportional sharing of shortages with no trading produces the greatest loss in farm income for adapting to shortage. These findings suggest that water trading arrangements have an important role play in adapting to water supply changes. Our results provide science-informed guidance to water managers, stakeholders, and policymakers tasked with discovering efficient water shortage adaptation plans for arid and semi-arid irrigation regions internationally.

1. Background

1.1. Introduction

Irrigated agriculture is significantly more productive where sufficient water is available than non-irrigated agriculture, particularly in the western U.S. Most of western U.S. irrigation access experienced large increased crop production over the 97 year period from 1910 to 2007, adding an estimated $19 billion to agricultural production in sales value (Edwards and Smith, 2018). In the USDA's 2022 Cash Rents Survey, the rental price of an acre of irrigated cropland in western states of the U.S. was an average of 3.65 times higher than the value of an acre of non-irrigated cropland. In 2017, fully irrigated farms in the western U.S. account for 31 percent of cropland acreage but 77 percent of the value of crop sales. Nationwide, these farms account for 8 percent of cropland and 30 percent of the value of crop sales (U.S. Department of Agriculture National Agricultural Statistics Service, 2019).

Agriculture in much of the arid western U.S. is economically infeasible without irrigation, for which irrigation can require transport of surface water resources across often considerable distances (Coman, 2011). Arid farmland in the western U.S., much of which was developed as a result of the 1862 Homestead Act, Desert Land Act of 1877, and Reclamation Act of 1902, as well as subsequent legislation, had too little water resources for profitable agriculture without irrigation (Allen, 1991, 2019; Libecap and Hansen, 2002; Libecap, 2011). Irrigation organizations, for which one example is the acequia organization seen in New Mexico (Freeman, 2000; Fernald et al., 2007, 2012; Cox and Ross, 2011; Cox, 2014), encapsulate an array of institutions that were

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developed in the late nineteenth and early twentieth centuries to address the collective action problem of constructing the infrastructure required to convey water to farmland (Bretsen and Hill, 2006).

Where cohesive communities existed, less formal organizations such as mutual ditch companies and acequias (Cox and Ross, 2011) served to pool resources to build needed diversion and conveyance infrastructure (Ostrom and Gardner, 1993; Ostrom, 2011). In some cases, private organizations or formal contracts facilitated the construction of this infrastructure (Leonard and Libecap, 2019). However, many of the largest irrigation organizations developed as part of the establishment of large public water projects by the U.S. Bureau of Reclamation (Doremus and Tarlock, 2003) or other federal and state entities (Pisani, 2002; Wahl, 2013). Investments in irrigation infrastructure produced considerable economic benefit, especially in times of drought (Hansen et al., 2011). In most of the western United States, the allocation of water during shortages is governed by the prior appropriations doctrine (Bluemel, 2004; Dridi and Khanna, 2005; Schlager, 2006) – in which priority is assigned to water rights that were established first (Tarlock, 2001). This priority is often capitalized into the value of a water right (De Mouche et al., 2011), with more senior water rights generally commanding a higher price (Lee et al., 2020). In practice, actual curtailments do not always occur, with the doctrine sometimes implemented to encourage cooperation instead of costly litigation (Tarlock, 2000). In addition, within irrigation districts, water shortages may affect more junior users first, or may be shared proportionally among all users by reducing deliveries through effectively implementing a proportional sharing of shortages (Ward et al., 2013; Salman et al., 2014, 2017; Gohar et al., 2015) alternating delivery days, or otherwise proportionally reducing total water deliveries. While water rights are generally subject to prior appropriations in the U.S. west, federal and state water projects may determine allocations based on factors including contracts, legal settlements, reservoir levels, and water stored in snowpack. For example, during the 2012–2016 drought in California, some irrigators received no water while others received their full allotment (Sugg, 2018). Federal and state decisions about water allocations can significantly impact water available for irrigation organizations.

In contrast, much of the midwestern and southeastern U.S., access to groundwater and adequate precipitation made the conveyance of water over long distances to farmland uneconomical, and surface water was generally governed under a system of riparian rights (Hu and Eheart, 2014) in which water users along rivers share shortages. Irrigation organizations in these areas are more focused on coordinating groundwater extraction and managing hydroelectric infrastructure with some attached irrigation infrastructure (Hrozencik et al., 2023).

The western U.S. experienced an unusually wet period in the early 1900s during which much of irrigated agriculture and accompanying irrigation organizations were being established (Woodhouse et al., 2005). The Colorado River Compact was signed in 1922, allocating over 20 million acre-feet of water when river flows were estimated earlier at 17.3 million (La Rue, 1916). This Compact signing occurred just seven years after 1915, the highlight of the wettest epoch since at least 800AD. During the past several decades, drought prevalence in much of the western U.S. has steadily increased (Cook et al., 2007). Demographic trends have also increased demand for water (Olmstead, 2010).

Changing climate has had a significant effect on reliable water supplies, particularly on water stored as snowpack. Higher temperatures and more variable precipitation reduce the level of water stored in snowpack and alter the timing of spring runoff (Hamlet et al., 2005; Rauscher et al., 2008; Qin et al., 2020; Musselman et al., 2021; Hale et al., 2023). Reduced snowpack leads to less water availability during peak growing season demand (Mote et al., 2018). Snowpack is expected by some to decline by 25 percent by 2050, with implications for both surface water and groundwater resources (Siirila-Woodburn et al., 2021).

Farmers may mitigate some of the impacts of climate change by adapting to warmer temperatures and to less reliable water supplies (Ward, 2022). However, econometric estimates of the mean capacity for mitigation through adaptation to changing climate vary widely (Burke and Emerick, 2016; Gammans et al., 2017; Mérel and Gammans, 2021). Improving on-farm irrigation application efficiency is often proposed as a possible solution to water shortages (Jensen, 2007; Njuki and Bravo-Ureta, 2018). In some cases, investments in on-farm irrigation infrastructure can reduce water use per acre for a given crop (Howell, 2003; Knox et al., 2012; Levidow et al., 2014). Farmers may also respond to higher on-farm irrigation efficiencies by switching to higher valued and more water intensive crops, planting more intensively, or planting additional acreage, making the relationship between investments in measures to improve irrigation efficiency such as drip irrigation and net water savings at the basin scale unclear, particularly when land or water are not binding constrains (Ward and Pulido-Velazquez, 2008; Grafton et al., 2018; Perez-Blanco et al., 2021).

Water markets are often proposed as a method for reducing the overall economic losses incurred from water shortages, allowing users to move water toward a higher economic valued set of uses (Howe et al., 1986; Chong and Sunding, 2006). However, hydrological characteristics, possible third party damages, and regulatory frameworks contribute to significant transaction costs, inhibiting the development of water markets (Brewer et al., 2008; Grafton et al., 2011). Despite these hurdles, water markets continue to develop and become more efficient in many areas in the western United States (Brookshire et al., 2004; Brown, 2006; Schwabe et al., 2020). Permanent transfers have increased, though these are often to urban uses (Donohew, 2009). Clarifying water rights and allowing for water trading may lead to allocative efficiencies through increased water marketing, increasing the economic value of water use (Arellano-Gonzalez et al., 2021; Browne and Ji, 2023).

Water marketing is made more complex by interactions between individual farmers and their irrigation organizations, which manage much of the irrigation water storage and conveyance infrastructure and may hold water rights or contract water from the Bureau of Reclamation or other suppliers. Individual farmers within an irrigation organization may not easily be able to trade water with others outside of that irrigation organization. In the 2019 Survey of Irrigation Organizations, participation in water markets in the Southwest (Arizona, New Mexico, and Utah) was small, with only three percent of irrigation organizations reporting having engaged in leasing in the prior five years. Yet eight percent of these organizations reported water trading among users within the organization (Potter et al., 2023). However, the volume of water being traded by irrigation organizations was much higher than the volume of water being traded within irrigation organizations. This occurs because quasi-public entities like irrigation organizations are likely to make large but infrequent trades, relative to individual farmers within an organization that may more frequently trade smaller amounts of water. The relative impact of irrigation organizations and individual farmers on water markets presents an area that needs additional research.

While markets can help improve water allocation efficiency and address temporary shortages, the transfer of water rights to other uses or locations may have adverse effects on agricultural communities, especially for traditional and lower income farmers, and surrounding ecosystems (Bajaj et al., 2022). It is well-known that permanent transfers and long term leases of water to other regions remove water from the community, increasing the burden of maintaining infrastructure by remaining community members (Edwards et al., 2008). Additionally, markets may reduce incentives to invest in more efficient irrigation infrastructure and farmers often wait until an event such as a drought significantly increases the value of water (Carey and Zilberman, 2002).

As coordinating entities charged with providing water to their constituent users or making implementing beneficial use of water, irrigation organizations stand to play an important role in water marketing. However, irrigation organizations may face incentives that lead them to under-participate in markets relative to private water rights holders

(Ghimire and Griffin, 2014). Additionally, prohibitions on transfers from agricultural to urban uses can impede water market participation by irrigation organizations (Grafton et al., 2012).

Despite these challenges, the infrastructure and organizational scales of irrigation organizations may make them well suited to participate in water markets in a way that most benefits their constituent users. Average annual leases from 2009 to 2018 in Arizona and California accounted for four percent and two percent of water supply, respectively (Schwabe et al., 2020). By way of comparison, between 3 and 17 percent of irrigation organizations in regions of the western U.S. engaged in water leasing from 2015 to 2019, which was between 1 and 9 percent of these organization's water supplies (Potter et al., 2023). This suggests both that irrigation organizations already account for a notable share of water volume traded, and that there may be additional gains from water marketing by these organizations for handling periods of water shortage.

1.2. Previous work

A number of earlier works has identified several drought and related water risk management plans (Keshavarz and Karami, 2016). Some existing works have looked into research and policy measures to elevate crop yields to deal with impacts of water shortages (Mishra et al., 2015; Murthy et al., 2015; Kulkarni et al., 2020), limit water depletions in irrigated agriculture (Blanco and Gomez, 2014); as well as protect access to affordable food prices (Hazell and Hess, 2010). Other works have investigated adjustments to various potential future water supplies (Tran et al., 2019; Chiambretto and Martin, 2020; Jovanovic et al., 2020; Serbouh et al., 2022; Zolfagharipoor et al., 2022; Cerasoli and Porporato, 2023; Han et al., 2023; O'Donnell et al., 2023) in addition to better drought management planning delivered by establishing workable water markets (Wei et al., 2011; Chengot et al., 2021).

Several previously published works have looked into measures to elevate crop yield while protecting sustainable water use (de Vito et al., 2017) practices in the face of water shortages (Assouline et al., 2015). Some work has assessed the issue of yield gaps (Godfray et al., 2010) to discover better crop planning strategies (Agus et al., 2019). Other papers have investigated predictors of crop yields, discovering that policies guided by discovering yield increasing methods can address the challenge by keeping tracking spatial elements of gaps in yields. A notable 2022 work analyzed yield gaps by looking at technical improvements (Zhi et al., 2022). Other works have found that in spite of attempts to elevate yields to protect food security, crop yield elevations have fallen off because many locations have already achieved maximum attainable yields in light of current technical limitations (Arata et al., 2020).

Additional research has investigated methods to conserve water use in agriculture when faced by water stress (Herath et al., 2013). Some point to the economic viability of certain water-conserving technologies which decreases water applied compared to conventional agriculture (Zotarelli et al., 2009). Drip irrigation has been found in some work to reduce water applications more than flood irrigation (Fonteyne et al., 2021). Yet, while drip irrigation can reduce water applications per acre for a given crop, the resulting increased marginal value of water can incentivize farmers to increase both water and land use and plant more water-intensive crops, which can actually increase crop ET and water depletions at the basin scale (Ward and Pulido-Velazquez, 2008; Grafton et al., 2018; Perez-Blanco et al., 2021).

Market incentives implemented by trading bulk water or water rights (Bouma et al., 2011) have been found to be a productive method to allocate water shortages in arid regions for climate adaptation (Kiem, 2013). Water trading (Montilla-Lopez et al., 2018) has also been shown to increase cropland efficiency with minimal loss of water's agricultural economic performance even when moving small amounts of water to urban uses (Gohar and Ward, 2010). A number of implementation issues can cause water markets to be less efficient than intended because of asymmetric information (Heaney et al., 2006). However, there is also evidence that even in the face of asymmetric information, there can still remain sufficiently strong trade incentives that allow for an increase in farm income. (Dridi and Khanna, 2005).

1.3. Gaps

The journal *Agricultural Water Management* has published a number of well-recognized works discovering a series of innovative policy measures to achieve more food value from a reduced amount of water brought on by drought or climate (Pereira et al., 2002; Ali et al., 2007; Bjornlund et al., 2009; Geerts et al., 2010; Ziolkowska, 2015; Molle and Tanouti, 2017; Ahmed, 2020). Several more recent works have also investigated policy measures for handling water shortages (Liu et al., 2024; Niyonsenga et al., 2024; Ongoma et al., 2024; Schraw et al., 2024; Zobeidi et al., 2024).

Despite these accomplishments, few peer-reviewed works in that or other journals we found presented research that conceptualizes, formulates, constructs, and applies an integrated optimization framework to inform debates on the design of adaptive water trading measures for handling water shortages. Some work of this kind has been done. For example, in the case of Spain, most trading is temporary and conditional on the formal approval of the government, which has historically occurred during droughts—thus, most research there focuses on temporary trading during shortages (Calatrava and Garrido, 2005; Pérez-Blanco et al., 2015, 2020; Rey et al., 2016; Calatrava and Martínez-Granados, 2018). More recent work has focused on how to design such temporary markets to maximize economic surplus and minimize budgetary pressures from a buyback perspective (Gómez-Limón et al., 2020; Gutiérrez-Martín et al., 2020; Sapino et al., 2023). So, there is some evidence that the issue of how exploring water sharing institutions that reduce economic and environmental damages during water shortages has been investigated. This work's comparison of alternative water sharing agreements is more innovative, although there is also some previous work on this, e.g., (Rey et al., 2016).

Few published works have integrated as a single framework a package of interacting parts of hydrologic science, agronomic research, economics, and policy analysis into a single framework to inform plans to systematically guide the path to affordable climate adaptation. Fewer still have integrated a system into a workable model that can be used to scientifically assess innovative policy measures connected to several elements, for which these connections take place over several locations, water trading arrangements, and water shortage scenarios.

1.4. Contribution

The original and novel contribution of this work is to address the gaps described above by answering this research question: "What innovative water sharing institutions can be designed and applied for irrigated agriculture to reduce economic damages that would otherwise occur from adapting to water shortages?" Its novel contribution is to address that question posed by conceptualizing, formulating, and applying an innovative optimization framework that discovers alternative water sharing methods based on their impact on farm income losses that would otherwise be incurred from water shortages.

This paper implements its contribution by formulating an integrated optimization model that integrates spatially and agronomically diverse cropping patterns with alternative water shortage sharing methods to discover water shortage adaptations over crops and locations that light a path to a plan that reduces the scale of farm income lost. The capacity supplied by this work's innovating contribution is to show how increased flexibility of water shortage sharing institutions can inform farmers and policymakers in arid regions internationally to more efficiently adapt to water shortages (Li et al., 2000; Ward and Pulido-Velazquez, 2008; Olmstead, 2014; Hunecke et al., 2017; Partey et al., 2018).

2. Methods of analysis

2.1. Overview

Our work formulates a nonlinear programming model for which optimized results can be used to inform discovery of economically efficient water shortage sharing plans when facing water shortage in a region of the western U.S. that has experienced many years of periodic water shortages (Ward et al., 2006; Margolis et al., 2007; Ward and Pulido-Velazquez, 2008; Lehner et al., 2017; MacEwan et al., 2017). As described in more detail below, our optimization model is calibrated to replicate historical land use by crop and county in the Middle Rio Grande Basin study area (Fig. 1, located at the end of the manuscript). By calibrating to historical land and water use patterns, our method permits a more efficient economic assessment of water shortage sharing institutions while adapting to future water supply stresses for which those future supplies can be expected to change from the historical period of record.

2.2. Study area

2.2.1. Farming regions

We investigate the Elephant Butte Irrigation District (EBID) and Middle Rio Grande Conservancy District (MRGCD) along with their associated six counties (Fig. 1). These two districts were selected because (1) each represents a unique economic motivation for irrigation activities, (2) each has different levels of hydraulic security in the face of highly streamflow fluctuations of the Rio Grande, and (3) and water users in all counties compete for water in the same river system. The MRGCD includes Socorro, Valencia, Bernalillo, and Sandoval Counties, while EBID includes Dona Ana and Sierra Counties (Fig. 1).

Counties practicing irrigation farming in this part of the Rio Grande Basin are located in an arid climate that has always been susceptible to drought (Dubois, 2020; U.S. U.S. U.S. National Vegetation Classification, 2020). The counties served by EBID are located in the Chihuahuan desert (Reynolds et al., 1999; Kurc and Small, 2004; Laliberte et al., 2004; Ludwig et al., 2005; Thomey et al., 2011) through which Rio Grande flows pass. Counties served by MRGCD are partly in the Chihuahua desert but also occur in the higher elevation dry regions of the northern part of New Mexico (Kinzli et al., 2015).

2.2.2. Cropping patterns and time frame

Cropping pattern data sourced from the U.S.D.A. Crop Layers Database were used to identify acreages of the most important crops in these six counties (U.S. Department of Agriculture, 2022b) for which importance is defined as total irrigated acreage. As shown in Table 1, the most important seven crops in these six counties are pecans, alfalfa, corn, wheat, peppers, cotton, and onions. We selected these crops because they make up more 90 percent of the acreage of all six counties. The year selected for this analysis is the most recent year from which we had reliable data, 2022. All tables are located at the end of the manuscript.

2.2.3. Water supply and institutional design

An essential policy-informing mission of this work is to assess a number of implementations of water trading for adapting to shortages. Three implementations of water trading are assessed: (1) unrestricted trading, defined as trading over all crops and all counties; (2) restricted trading, defined as trading among all crops within each county, but no trading across county lines, for which the administrative costs are lower than unrestricted trading; and (3) no trading at all, for which no trade is

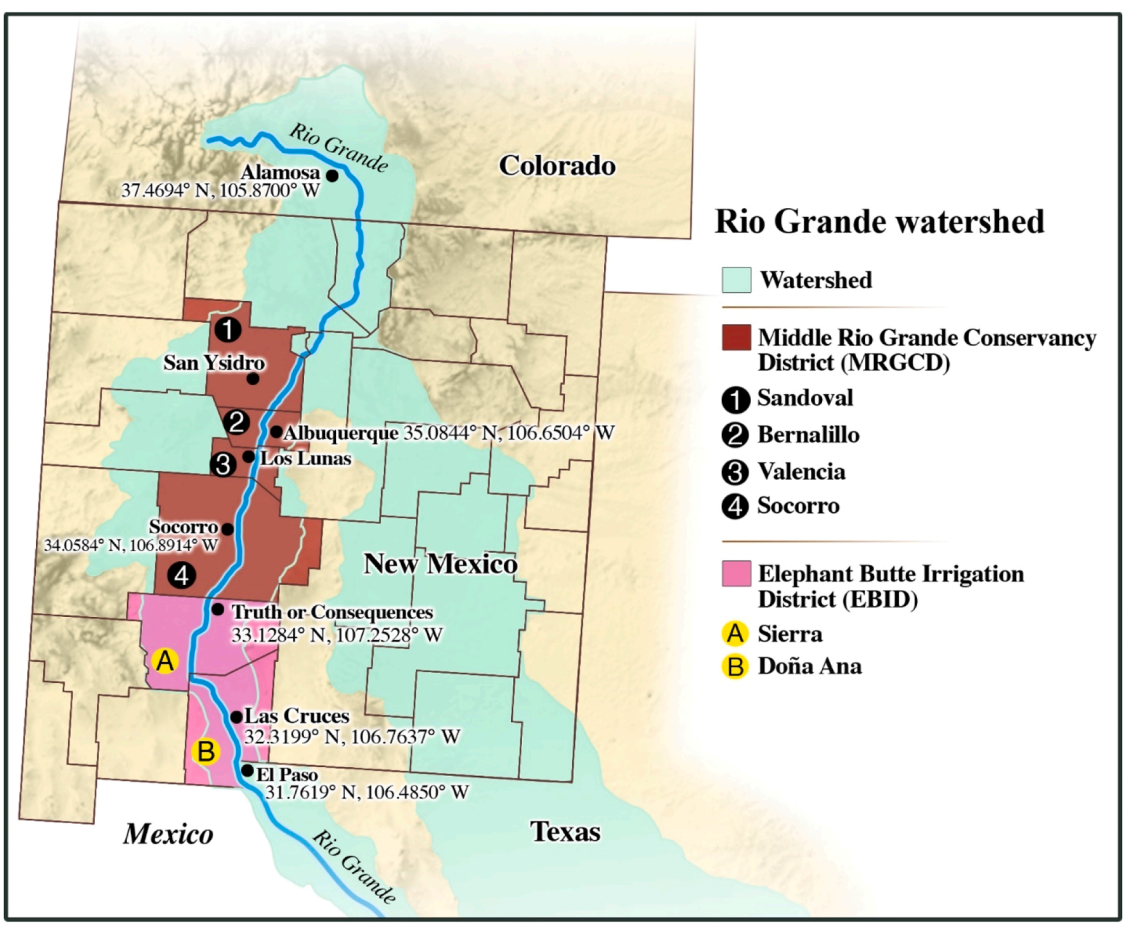

Fig. 1. Rio Grande Watershed, North America.

Table 1
Total observed land in production by crop, county, and irrigation District, 2022 (acres).

District	Elephant Butte Irrigation District		Middle Rio Grande Conservancy District			
Crop/County	Dona_Ana	Sierra	Socorro	Valencia	Bernalillo	Sandoval
1_pecan	39,372.3	977.2	426.3	731.2	78.1	61.6
2_alfalfa	13,074.1	2052.3	9051.7	7010.8	517.5	1293.9
3_corn	662.3	501.9	463.0	340.5	18.7	55.8
4_wheat	1251.2	576.0	151.4	153.0	34.7	66.0
5_peppers	1498.3	572.2	212.6	10.5	2.0	16.7
6_cotton	14,283.1	898.5	265.1	46.0	5.8	8.0
7_onions	1988.7	568.4	133.2	188.4	5.6	12.0
Total	72,130.0	6146.5	10,703.3	8480.4	662.4	1514.0

allowed to adapt to water shortages when those shortages occur. For the third method of handling shortages, there is a simple proportional sharing of shortages, for which we specify an equal proportion of the overall shortage to each crop and county, as that shortage sharing method requires comparatively little administrative machinery (Ward et al., 2013; Salman et al., 2014; Gohar et al., 2015).

Two water supply situations were assessed for the present work. The two consist of (1) a base historical water supply for which no major drought stress occurs, as well as (2) a scenario under a water shortage, defined arbitrarily here as 75 percent of the total water applied to irrigation under the full supply. A large number of other water shortage scenarios could have been presented, but space limitations constrain performing this assessment, despite there being potentially considerable interest. We selected the Rio Grande watershed (Fig. 1) for purposes of illustration, but the methods developed and applied could see use in a number of other watersheds domestically or even internationally. The complete model (GAMS) code and spreadsheet output are included as Appendices A and B.

2.3. Data

2.3.1. Land

Irrigated land observed for this model is based on total acreage for each of the seven crops described earlier for 2022. Land data were derived from the USDA Crop Layer Database (CDL), and cropped acres for each crop were aggregated to produce county totals for each county. Sierra and Dona Ana County acreage falls under the EBID, while land in Socorro, Valencia, Bernalillo, and Sandoval County land in production are in MRGCD (U.S. Department of Agriculture, 2022c).

2.3.2. Crop yields

Crop yields are sourced from the state's most recent agricultural statistics, for which all yields were converted to US tons (1 US ton = 0.907 metric tons) per acre (U.S. Department of Agriculture, 2022a). Irrigation in counties served MRGCD typically produces lower crop yields compared to counties served by EBID, based on enterprise budgets published by the regional land grant university (New Mexico State University, 2022). Crop yields were adjusted from regional averages for those MRGCD county irrigated crops not included in the university's crop budgets.

2.3.3. Crop prices

Crop prices varied by crop but not by county. The most recent data on crop prices came from 2022 state agricultural statistics. (U.S. Department of Agriculture, 2022a). Crop prices show minimal variability across counties, for crops are typically traded in national and in some cases international markets.

2.3.4. Crop production costs

Costs of production come from the regional land grant university published cost and return budgets (New Mexico State University, 2022). As published the costs per are shown for a typical farm with above-average management. Where production costs per acre are not available certain crops such as pecans, estimates are set to 70 % of gross revenue per acre. The cost of water is included in the enterprise budgets.

2.3.5. Crop water use (ET)

Crop water use (ET) information is secured from satellite data. These data come from a new data base system developed in 2022 known as OpenET (OpenET, 2023). This comparatively new data source was publicized widely in 2023, after a development period of several years. Still, as of late 2023, the OpenET data are still available only for the western part of the U.S. Those satellite derived data are based on the use of well-established methods to generate monthly and annual ET estimates at the field scale. Units defining water use for OpenET for that database include both water depth and water volume. Volume is defined algebraically as depth multiplied by acreage of the field. In its current development, Open ET does not distinguish water sourced from surface versus groundwater sources (OpenET, 2023). Users of this OPEN ET database can assess ET data at the field scale for millions of individual farming fields based on satellite data down to a field scale of one-quarter of an acre resolution (one acre = 0.405 ha).

2.4. Model description

2.4.1. Mathematical programming model

Our work is a mathematical programming optimization model. It uses what has become known in recent years is known as positive mathematical programming, or PMP (Mirra et al., 2023). It grows from an earlier work developed by Howitt (1995) and a number of works emerging from that original. It implements principles described in more detail elsewhere (Dagnino and Ward, 2012). Our model uses the well-known first order necessary conditions for optimizing farm income with respect to land and irrigation water use to derive parameters that reflect crop yield as a linear function of the scale of land in production (Iglesias and Blanco, 2008; Cortignani and Severini, 2009; Medellín-Azuara et al., 2010; Howitt et al., 2012).

2.4.2. Model calibration

This work mathematically specifies a linear yield (Y) function for each crop in equation (1) below. That equation shows a decreasing marginal yield for each crop associated with higher levels of irrigated land entering production. The level of land in production (Land), by crop (i = 1…7) and county (j = 1…6) shows diminishing marginal yields for brought on by additional land entering production. This mathematical approach is designed to guard against farmers completely specializing in any single crop at the county level under observed conditions, since for our crops chosen, no complete specialization occurred. However, under a water shortage condition, the lowest valued crops can fall out of production when there is too little water to economically justify keeping it in production. It is common to see an individual farm manager specializing in a single crop, especially for smaller acreages. However, at the county level, complete specialization is rarely seen, and is not seen at all for any of our six counties investigated.

We developed a crop water production function to account for any given crop's yield declining as additional land allocated to that crop. Falling crop yields in the face of additional production reveal the principle of Ricardian rent (Souty et al., 2012) in which each crop's highest yielding lands enter production first, upon which crop yields are lower as less productive land is brought into production attracted by higher crop prices (Cairns, 1986; Tilton, 2004; Kratena, 2008; Gholizadeh et al., 2021). Along the lines of an earlier development (Dagnino and Ward, 2012) the present work specified then estimated from observed data a crop yield model to account for land of varying quality. For those lands, crop yield (*Y*) fall with greater amounts acreage (*Land*) allocated to a crop, with more marginal lands coming into production with expanded acreage:

$$Y_{ij} = \alpha_{0ij} + \alpha_{1ij}\,Land_{ij}$$

The term α_0 reflects the crop yield for the first unit of land entering production, while the term α_1 reflects the incremental effect of additional land on average crop yield. For each crop (i) and county (j), the quantity of water used per unit land is fixed, which accounts for water used in irrigation per unit of land irrigated. So total water use for each i-th crop planted in each j-th county is measured by total land assigned to that crop multiplied by the crop water applied per unit land.

The calibration analysis described above is achieved by combining observed facts with an underlying theory that must have generated those facts. Its application results in the calculation of the previously unknown terms α_0 and α_1 described in equation (1). The α_0 and α_1 terms were calculated as those which if used by farmers, would have exactly reproduced observed acreage of each crop in each district. Additional details along with a mathematical derivation appear in an earlier work (Dagnino and Ward, 2012). After calculating these parameters, running a non-constrained optimization using the calculated values of the parameters α_0 and α_1, can be expected to produce observed land, water, and income for each i-th crop and j-th county that matches observations, an efficient model calibration approach, as described in an earlier work (Gohar et al., 2015). Overall, the process described here carries out two steps described below:

- Observed data: We measure crop yield, income, land irrigated, and water use for by crop and county.
- Economic theory: When irrigators make choices that maximize net income based on a quadratic objective income function because of the linear yield term described in (1), growers at the county level expand water applied from zero to the optimized level at which the value of the marginal product of water equals its price.

The yield parameters, α_0 and α_1, described in equation (1) are placed into the correct spots for each crop's yield function for each county. As described in more detail in an earlier work (Gohar et al., 2015), the results reveal results of an optimized total farm income at the regional level earned from water applied to each crop. After the calibration occurs, the optimization model produces results that require no water constraints whatsoever. For handling drought, upper bounds on water use are required to handle whatever shortage is specified.

3. Results

3.1. Observed data

3.1.1. Land

Table 1 shows observed values of land in production by crop for each of our six Rio Grande Basin New Mexico Counties. Results shown in the table all have an influence on the economic efficiency of water trading institutions that could be implemented to adapt to drought and climate change in the western United States. Several patterns are noted in the table:

- Comparatively large irrigated acreage in Dona Ana County (72 percent of all counties).
- Comparatively large proportion of Dona Ana County acreage in pecan orchards (54 percent of the total acreage). This is not surprising as pecans have a long and established track record of being a high valued crop in the region.
- The large acreage of high valued pecans suggest that when unrestricted water trading is permitted as a drought adaptation measure, much of that water would flow to pecan acreage.

3.1.2. Irrigation water depth

Table 2 shows observed water depth by crop, county, and irrigation district for the year 2022, based on data from the newly-developed OPEN ET database (US Geological Survey, 2023) A few interesting patterns emerge from this table:

Lower water use per irrigated acre (ET) occurs in moving geographically from south to north. This movement occurs with elevations of the Rio Grande rising from about 3880 feet to 5200 feet above sea level, for which latitude increases from 32.3199° N to 35.5148° N. Pecans tend to have the highest ET rate among the basin's most widely grown irrigated crops, while cotton has the lowest. Greater ET conserves the lowest amount of water, but also can have high planted acreage, mostly because the highest ET crops also produce the highest farm income per acre or per unit of water used. High acreage levels coinciding with high ET point to the central importance of net income per unit water use as a driver of economic activity.

3.1.3. Net revenue per acre

Table 3 shows observed net revenue per acre by crop, county and irrigation district in dollars per acre. Net revenue per acre is defined as price multiplied by yield minus costs of production. The table shows wide swings in net revenue per acre ranging from a low for wheat equal to $11.55 per acre in Socorro County to a high of $2610.97 per acre for chile peppers in Dona Ana County. We would expect marginal values per acre foot of water to be equal since irrigation farmers in the region are typically income maximizers, this maximization does not produce equal net income per acre, as shown in the table.

3.2. Optimized results

3.2.1. Total water use

Table 3a shows results of water use from a series of optimization model runs. It shows results for two water supply scenarios. Full water use is defined as total observed crop-water ET by irrigated agriculture for the year 2022, while drought use refers to a reduction in crop water to the extent of 75 % of actual observed value. We had no reliable data on water applied, diverted, or pumped. This table is merely an accounting table, designed to set the stage for several follow-up tables showing crop-water use (ET) for three water trading arrangements under each of the two drought scenarios considered.

3.2.2. Land in production

Table 4 shows a number of notable patterns: It shows complete elimination of acreage for the lowest valued crops (wheat) in all counties when drought occurs for both an unlimited and limited scale of water trading, but only a 25 percent acreage reduction for that crop when no trading is permitted. It also shows a modest reduction of acreage in high valued crops like pecans, peppers, and onions for both an unlimited as well as limited scale of water trading, but an identical 25 percent acreage reduction across all crops when no trading is permitted. Finally, it shows that when the geographical scale of water trading is larger, a higher proportion of acreage is maintained in the highest valued crops (pecans, peppers, onions) than when the scale of trading is more limited. The high proportion occurs because there is greater economic opportunity to move water longer distance to secure higher income for the highest valued crops.

Table 2
Observed Water Depth by Crop, County, and Irrigation District, 2022 (feet).

District	Elephant Butte Irrigation District		Middle Rio Grande Conservancy District			
crop/county	Dona_Ana	Sierra	Socorro	Valencia	Bernalillo	Sandoval
1_pecan	4.50	4.00	3.75	3.58	3.51	3.40
2_alfalfa	3.42	3.17	2.83	2.75	2.67	2.58
3_corn	2.92	2.83	2.50	2.33	2.28	2.17
4_wheat	3.67	3.50	3.33	3.00	2.86	2.77
5_peppers	2.83	2.67	2.33	2.28	2.21	2.14
6_cotton	2.50	2.33	2.07	2.01	1.95	1.89
7_onions	3.00	2.83	2.49	2.41	2.34	2.27
Unweighted average	3.26	3.05	2.76	2.62	2.55	2.46

Table 3
Observed Net Revenue Per Acre by Crop, County, and Irrigation District for 2022 ($ US).

crop/county	Irrigation District					
	Elephant Butte Irrigation District		Middle Rio Grande Conservancy District			
	Dona_Ana	Sierra	Socorro	Valencia	Bernalillo	Sandoval
1_pecan	866.00	838.52	803.84	510.37	620.67	571.47
2_alfalfa	595.69	568.37	488.58	483.43	457.41	458.43
3_corn	409.40	400.18	389.60	324.51	307.19	290.18
4_wheat	25.89	32.03	11.55	24.91	26.58	11.56
5_peppers	2610.97	2496.00	2143.21	2097.33	2096.69	2069.06
6_cotton	135.05	70.52	69.61	52.07	67.53	62.15
7_onions	2398.46	2259.57	2014.24	1943.53	1886.76	1809.77

Table 3a
Total Water Use by Water Supply, and Water Trading Institution, Summed Over Six Rio Grande Basin Counties, 2022 (acre-ft/year).

Water Supply	Water Trading Scope		
	unlimited	limited	none
full	353,131	353,131	353,131
drought	264,848	264,848	264,848

Our model at its current stage of development does not allow for responding to water shortages through intensive margin factors like deficit irrigation. Especially in the case of perennial crops like pecans, adjustments on the intensive margin are likely to reduce estimated impacts of water shortages on adjustments on the extensive margin, namely land in production. The estimated land in production for pecans in Table 4 under drought scenarios should therefore be taken as a lower bound. In other words, land in production indicated in Table 4 is at this level with no intensive margin adjustments, and would be higher if those adjustments were modelled.

3.2.3. *Water use*

Table 5 shows several important patterns in connection with water use. The total observed water use of 353,131 acre feet used over the basin's seven highest acreage crops (not shown) matches the optimized level under base conditions, reflecting the proper working of the PMP model calibration method. Total drought-induced water use is 264,848 acre feet for the year 2022, reflecting the assumed shortage level of 25 percent of full supplies. The shortage induced level of water is equal for all three water trading arrangements assessed.

It also shows that with unlimited trading, water traded moves to its highest valued use, pecans, peppers, and onions. These three crops, respectively maintain 82 percent of full supply use, 96 percent, and 96 percent, reflecting their high economic value in maintaining nearly their full supply use, even when overall supplies are reduced 75 percent of full. The lowest valued crops are wheat (zero percent of full supply use) and cotton (33 percent of full supply use).

The table also shows that with limited trading, a similar pattern is shown, with much of the water gravitating to pecans, peppers, and onions. Wheat again keeps zero percent of its historical water use for all counties, reflecting its comparatively low profitability per unit water use. Finally, the table shows that when no trading occurs, all crops and all counties use the same 75 percent of fully supplied water. There is no allocation to any individual farm, but only to the total land and water allocation by crop and county. This proportional sharing of shortages can be seen as an equitable water shortage sharing plan, but even if it is equitable, it is not economically efficient for farm income or for food security, as the lack of market forces blocks water from moving out of low valued crops into higher valued ones.

3.2.4. *Crop yield*

Table 6 shows optimized crop yield (tons per acre) by district, crop, water supply scenario, and scope of trading. An important pattern seen in that table is that the smallest change in yield from historical levels is where the largest percentage of historical water use is maintained. For example, pecan yields in Dona Ana County rise from a historical 0.76 tons per acre to 0.80 tons per acre for the case of unlimited water trading because 82 percent of historical acreage is maintained in the unlimited water trading scenario when facing drought. A similar pattern is shown for all crops and counties. The largest change in yields occurs for the smallest percentage of land maintained under the no trading scenario for the lowest values crops, which is shown in the table for the case of wheat for all counties. For the linear yield function specified larger reductions in the scale of production raise yields.

3.2.5. *Farm Income*

Table 7 shows optimized total farm income by irrigation district, water supply scenario, and water trading institution. This table demonstrates the important principle that greater amounts of geographic scope in water trading allowed protects a larger proportion of base farm income. When unrestricted trading is implemented, 95.59 percent of

Table 4
Optimized Land in Production by Crop, New Mexico Rio Grande Basin County, Water Supply Scenario, and Scope of Water Trading Allowed, 2022 (acres and proportion of full).

crop	county	Water Supply Scenario						
		full	drought					
			Water Trading Scope					
			unlimited	prop	limited	prop	none	prop
1_pecan	Dona_Ana	39372.3	32348.0	0.822	32348.4	0.822	29,529.2	0.75
1_pecan	Sierra	977.2	817.2	0.836	884.1	0.905	732.9	0.75
1_pecan	Socorro	426.3	358.0	0.840	347.5	0.815	319.7	0.75
1_pecan	Valencia	731.2	554.9	0.759	525.7	0.719	548.4	0.75
1_pecan	Bernalillo	78.1	62.9	0.806	62.7	0.803	58.6	0.75
1_pecan	Sandoval	61.6	49.0	0.796	47.9	0.778	46.2	0.75
2_alfalfa	Dona_Ana	13074.1	10499.5	0.803	10499.7	0.803	9805.6	0.75
2_alfalfa	Sierra	2052.3	1659.7	0.809	1824.0	0.889	1539.2	0.75
2_alfalfa	Socorro	9051.7	7249.5	0.801	6973.0	0.770	6788.8	0.75
2_alfalfa	Valencia	7010.8	5641.6	0.805	5414.2	0.772	5258.1	0.75
2_alfalfa	Bernalillo	517.5	413.9	0.800	412.5	0.797	388.1	0.75
2_alfalfa	Sandoval	1293.9	1043.6	0.807	1021.9	0.790	970.4	0.75
3_corn	Dona_Ana	662.3	500.3	0.755	500.3	0.755	496.7	0.75
3_corn	Sierra	501.9	379.9	0.757	431.0	0.859	376.4	0.75
3_corn	Socorro	463.0	361.0	0.780	345.3	0.746	347.3	0.75
3_corn	Valencia	340.5	256.4	0.753	242.5	0.712	255.4	0.75
3_corn	Bernalillo	18.7	13.9	0.746	13.9	0.742	14.0	0.75
3_corn	Sandoval	55.8	41.5	0.744	40.3	0.721	41.9	0.75
4_wheat	Dona_Ana	1251.2	0.0	0.000	0.0	0.000	938.4	0.75
4_wheat	Sierra	576.0	0.0	0.000	0.0	0.000	432.0	0.75
4_wheat	Socorro	151.4	0.0	0.000	0.0	0.000	113.6	0.75
4_wheat	Valencia	153.0	0.0	0.000	0.0	0.000	114.8	0.75
4_wheat	Bernalillo	34.7	0.0	0.000	0.0	0.000	26.0	0.75
4_wheat	Sandoval	66.0	0.0	0.000	0.0	0.000	49.5	0.75
5_peppers	Dona_Ana	1498.3	1442.5	0.963	1442.5	0.963	1123.7	0.75
5_peppers	Sierra	572.2	551.2	0.963	560.0	0.979	429.2	0.75
5_peppers	Socorro	212.6	204.7	0.963	203.4	0.957	159.5	0.75
5_peppers	Valencia	10.5	10.1	0.963	10.0	0.956	7.9	0.75
5_peppers	Bernalillo	2.0	1.9	0.964	1.9	0.963	1.5	0.75
5_peppers	Sandoval	16.7	16.1	0.964	16.1	0.961	12.5	0.75
6_cotton	Dona_Ana	14283.1	5205.6	0.364	5206.0	0.364	10,712.3	0.75
6_cotton	Sierra	898.5	0.0	0.000	305.0	0.339	673.9	0.75
6_cotton	Socorro	265.1	0.0	0.000	0.0	0.000	198.8	0.75
6_cotton	Valencia	46.0	0.0	0.000	0.0	0.000	34.5	0.75
6_cotton	Bernalillo	5.8	0.0	0.008	0.0	0.000	4.3	0.75
6_cotton	Sandoval	8.0	0.0	0.000	0.0	0.000	6.0	0.75
7_onions	Dona_Ana	1988.7	1903.3	0.957	1903.3	0.957	1491.5	0.75
7_onions	Sierra	568.4	543.9	0.957	554.2	0.975	426.3	0.75
7_onions	Socorro	133.2	127.6	0.958	126.7	0.951	99.9	0.75
7_onions	Valencia	188.4	180.4	0.957	179.0	0.950	141.3	0.75
7_onions	Bernalillo	5.6	5.4	0.957	5.4	0.957	4.2	0.75
7_onions	Sandoval	12.0	11.5	0.957	11.4	0.953	9.0	0.75

base income is maintained, while smaller based levels are maintained for the two more restricted water trading levels allowed. In fact the loss in farm income ($4.3 million) is about 1.5 times as high under no trading as under unrestricted trading ($2.8 million).

3.2.6. *Economic value of additional water*

3.2.6.1. No shortage. Table 8 shows the optimized economic value of additional water by New Mexico Rio Grande Basin county, crop, water supply scenario, and water trading institution, measured in $US per acre foot. These optimized values of incremental value are often termed 'shadow prices,' and they reflect the incremental economic value per acre foot of water consumed. For agricultural water use, the shadow price is the economic value associated with one more acre foot of water consumed, if made available for use in irrigated crops (D'Odorico et al., 2020). Where nominal or accounting water prices are zero, shadow prices provide a mathematically rigorous approach to adjust those zero accounting prices to reflect the income- generating power of an additional unit of water.

The shadow prices presented in this table can support cost-benefit analysis by assessing the economic merits of a program that would supply additional water for crop irrigation. The use of shadow prices to inform public water policy debates can support choices on the extent to which additional water should be made available for adapting to drought conditions. Use of the shadow prices presented in this table show that the economic impacts (gains) resulting from a water supply augmentation project. Shadow prices are zero when there are no shortages, for which results are shown in the first column of numerical results. That incremental value, namely zero, is equal for all crops and counties, reflecting no (additional) scarcity for any place or crop under observed base conditions. The table also shows how these shadow prices change by county, crop, and water trading arrangement, as described in detail below.

3.2.6.2. Shortage with unrestricted (Unlimited) trading. The shadow price of water is equal for all crops and county locations, at $68.67 per acre foot when unrestricted trading is allowed. Unrestricted trading would not be a logistically simple matter (Qureshi et al., 2007, 2012; Salman et al., 2014) because of the long river mile distances involved. Still, if the legal, delivery, and logistical system for long distance trading could be

Table 5
Optimized Water Use by Crop, New Mexico Rio Grande Basin County, Water Supply Scenario, Water Trading Institution, 2022 (acre-ft and proportion of full).

crop	county	Water Supply Scenario						
		full	drought					
		Water Trading Scope						
			unlimited	proportion full	limited	proportion full	none	proportion full
1_pecan	Dona_Ana	177,175	145,566	0.82	145,568	0.82	132,882	0.75
1_pecan	Sierra	3909	3269	0.84	3537	0.90	2932	0.75
1_pecan	Socorro	1599	1343	0.84	1303	0.82	1199	0.75
1_pecan	Valencia	2620	1989	0.76	1884	0.72	1965	0.75
1_pecan	Bernalillo	274	221	0.81	220	0.80	206	0.75
1_pecan	Sandoval	210	167	0.80	163	0.78	157	0.75
2_alfalfa	Dona_Ana	44,670	35,873	0.80	35,874	0.80	33,502	0.75
2_alfalfa	Sierra	6499	5256	0.81	5776	0.89	4874	0.75
2_alfalfa	Socorro	25,646	20,540	0.80	19,757	0.77	19,235	0.75
2_alfalfa	Valencia	19,280	15,514	0.80	14,889	0.77	14,460	0.75
2_alfalfa	Bernalillo	1380	1104	0.80	1100	0.80	1035	0.75
2_alfalfa	Sandoval	3343	2696	0.81	2640	0.79	2507	0.75
3_corn	Dona_Ana	1932	1459	0.76	1459	0.76	1449	0.75
3_corn	Sierra	1422	1076	0.76	1221	0.86	1067	0.75
3_corn	Socorro	1158	902	0.78	863	0.75	868	0.75
3_corn	Valencia	795	598	0.75	566	0.71	596	0.75
3_corn	Bernalillo	43	32	0.75	32	0.74	32	0.75
3_corn	Sandoval	121	90	0.74	87	0.72	91	0.75
4_wheat	Dona_Ana	4588	0	0.00	0	0.00	3441	0.75
4_wheat	Sierra	2016	0	0.00	0	0.00	1512	0.75
4_wheat	Socorro	505	0	0.00	0	0.00	379	0.75
4_wheat	Valencia	459	0	0.00	0	0.00	344	0.75
4_wheat	Bernalillo	99	0	0.00	0	0.00	74	0.75
4_wheat	Sandoval	183	0	0.00	0	0.00	137	0.75
5_peppers	Dona_Ana	4245	4087	0.96	4087	0.96	3184	0.75
5_peppers	Sierra	1526	1470	0.96	1493	0.98	1144	0.75
5_peppers	Socorro	496	478	0.96	475	0.96	372	0.75
5_peppers	Valencia	24	23	0.96	23	0.96	18	0.75
5_peppers	Bernalillo	4	4	0.96	4	0.96	3	0.75
5_peppers	Sandoval	36	35	0.96	34	0.96	27	0.75
6_cotton	Dona_Ana	35,708	13,014	0.36	13,015	0.36	26,781	0.75
6_cotton	Sierra	2096	0	0.00	712	0.34	1572	0.75
6_cotton	Socorro	550	0	0.00	0	0.00	412	0.75
6_cotton	Valencia	93	0	0.00	0	0.00	69	0.75
6_cotton	Bernalillo	11	0	0.01	0	0.00	8	0.75
6_cotton	Sandoval	15	0	0.00	0	0.00	11	0.75
7_onions	Dona_Ana	5966	5710	0.96	5710	0.96	4475	0.75
7_onions	Sierra	1610	1541	0.96	1570	0.97	1208	0.75
7_onions	Socorro	331	317	0.96	315	0.95	249	0.75
7_onions	Valencia	455	436	0.96	432	0.95	341	0.75
7_onions	Bernalillo	13	13	0.96	13	0.96	10	0.75
7_onions	Sandoval	27	26	0.96	26	0.95	20	0.75

established, the fact that water flows to higher economic values (money) makes that kind of water movement to equalize shadow prices across crops and locations a result that approximates the reality of such unrestricting trading if permitted and implemented. This kind of trading would occur entirely within a single US western state, New Mexico, for which trading would operate under a consistent and fairly unified set of water laws (Harding et al., 1995; Ward and Booker, 2003, 2006; Womble et al., 2022).

The principles of measurement, management, and marketing (Jorgensen and Syme, 2000; Hamlet et al., 2002; Sæle and Grande, 2011; Nguyen and Wagner, 2017) have been embraced by New Mexico's water planners for several years, so the table's results reflect the kinds of shadow prices that one might expect under the hydrologic and economic conditions shown in this paper. Future changes in crop prices, yields, or costs of production would alter the results presented in this table.

3.2.6.3. Shortage with limited trading. The middle column of optimized results under the drought scenario shows results of limited scope water trading, by which trading would be limited to trading water among only crops in the same county as observed for the year 2022. Trading would not be permitted across county lines for this scenario, in contrast to the first scenario for which unrestricted trading could occur among all crops and counties.

The table shows an equality of shadow prices across crops within each of the counties analyzed. For example, results of that column show the shadow price of water of $68.67 for all crops within Dona Ana County. This means that if total water supplies used in 2022 in Dona Ana County were reduced to 75 % of that observed use, then market forces would reallocate water among the various crops shown until the incremental value of one more acre foot applied is equal to $68.66. A more severe drought than that leaving 75 % available in Dona Ana County would produce a higher shadow price than the $68.66 shown in the table. A similar interpretation of these limited trading shadow prices is seen for all six counties shown in the table.

3.2.6.4. Shortage with no trading allowed. Notable patterns emerge for the case when no water trading is allowed, shown in the far right data column of Table 8. Shadow prices when no trading is allowed range from a low of $1.73 per acre foot (wheat in Socorro County) to a high of $482.91 per acre foot (onions in Sandoval County).

Table 6
Optimized Crop Yield Per Acre by Irrigation District, Crop, Water Supply Scenario, and Scope of Trading Permitted, 2022 (tons per acre).

Crop	County	Water Supply Scenario			
		full	drought		
			Water Trading Scope		
			unlimited	limited	none
1_pecan	Dona_Ana	0.76	0.80	0.80	0.82
1_pecan	Sierra	0.74	0.78	0.76	0.79
1_pecan	Socorro	0.71	0.74	0.75	0.76
1_pecan	Valencia	0.45	0.48	0.49	0.48
1_pecan	Bernalillo	0.55	0.58	0.58	0.59
1_pecan	Sandoval	0.50	0.53	0.54	0.54
2_alfalfa	Dona_Ana	4.76	5.16	5.16	5.27
2_alfalfa	Sierra	4.54	4.92	4.76	5.03
2_alfalfa	Socorro	3.90	4.24	4.29	4.32
2_alfalfa	Valencia	3.86	4.19	4.24	4.28
2_alfalfa	Bernalillo	3.66	3.97	3.97	4.05
2_alfalfa	Sandoval	3.66	3.97	3.99	4.06
3_corn	Dona_Ana	2.59	3.01	3.01	3.02
3_corn	Sierra	2.54	2.94	2.77	2.95
3_corn	Socorro	2.47	2.82	2.88	2.87
3_corn	Valencia	2.06	2.39	2.44	2.39
3_corn	Bernalillo	1.95	2.27	2.27	2.26
3_corn	Sandoval	1.84	2.14	2.17	2.14
4_wheat	Dona_Ana	0.31	0.41	0.41	0.34
4_wheat	Sierra	0.39	0.50	0.50	0.41
4_wheat	Socorro	0.14	0.18	0.18	0.15
4_wheat	Valencia	0.30	0.39	0.39	0.32
4_wheat	Bernalillo	0.32	0.42	0.42	0.34
4_wheat	Sandoval	0.14	0.18	0.18	0.15
5_peppers	Dona_Ana	6.40	6.51	6.51	7.15
5_peppers	Sierra	6.12	6.22	6.18	6.84
5_peppers	Socorro	5.25	5.35	5.36	5.87
5_peppers	Valencia	5.14	5.23	5.25	5.75
5_peppers	Bernalillo	5.14	5.23	5.23	5.74
5_peppers	Sandoval	5.07	5.16	5.16	5.67
6_cotton	Dona_Ana	0.65	0.69	0.69	0.67
6_cotton	Sierra	0.34	0.37	0.36	0.35
6_cotton	Socorro	0.34	0.36	0.36	0.34
6_cotton	Valencia	0.25	0.27	0.27	0.26
6_cotton	Bernalillo	0.33	0.35	0.35	0.33
6_cotton	Sandoval	0.30	0.33	0.33	0.31
7_onions	Dona_Ana	22.95	23.07	23.07	23.64
7_onions	Sierra	21.62	21.73	21.68	22.27
7_onions	Socorro	19.27	19.37	19.38	19.85
7_onions	Valencia	18.59	18.69	18.71	19.15
7_onions	Bernalillo	18.05	18.14	18.15	18.59
7_onions	Sandoval	17.32	17.40	17.41	17.83

Large variability is seen across crops and counties, reflecting widely varying economic values of an additional unit of water if made available under trading. High numbers reflect the highest economic scarcity, while low numerical values reflect low capacity to generate additional farm income. A high shadow price means that much value could be gained from the first unit of water taken out of a lower valued use and put into the use with the high shadow price. Examples are shown by the high shadow price of pecans, peppers, and onions for all counties. Similarly, low shadow prices provide a signal of comparatively low economic values displaced my moving water out of these crops, with a notable example being wheat. With unrestricted trading those low and high shadow prices would tend to move to the level of $68.67 shown in the far left data column under the drought scenario.

A numerical shadow price value higher than the overall average under unlimited trading means that water is scarcer for this crop and county than would occur if unlimited trading were allowed for handling the drought scenario. For example, the shadow price of $96.22 for pecans in Dona Ana County means that a proportional sharing of shortages with no trading allowed assigns less water to pecans in Dona Ana County (132,882 acre feet shown in Table 5) that would occur with unlimited trading (145,566 acre feet shown in the same table). A shadow price lower under no trading means the opposite, namely that more water and an associated lower marginal value is assigned with no trading allowed than with unlimited trading allowed. For example, allocating 75 % of historical water to wheat in Dona Ana County without trading produces $3.53 per acre foot shadow price (3441 acre feet), while with unlimited trading that same water allocated to wheat produces $68.67 in value (zero acre feet).

4. Discussion

4.1. Importance

This paper has analyzed the extent to which water trading can perform economically in arid and semi- arid climates under conditions of water shortages. The importance of this work is especially notable for policymakers, farm managers, or water planners who have a need to guide water shortage planning in irrigation dominated regions common in the American west facing the threat of current or potential water shortage episodes. Both the theoretical framework, methods used, and results found provide a resource of value for water stakeholders who need a set of tools to better guide programs or proposals for adapting to current or anticipated water stress. The theories, methods, and findings of this paper supply information that can guide farm managers in making better informed more economical decisions in a range of conditions. This work also provides a resource to water professionals who need more extensive data on water trading implementations and their

Table 7
Optimized Total Income by New Mexico Rio Grande Basin County, Water Supply Scenario, and Scope of Water Trading Permitted, 2022 ($ US and Proportion of Full Supply).

County	Water Supply						
	full	drought					
		Water Trading Scope					
		unlimited	proportion compared to full supply	limited	proportion compared to full supply	none	proportion compared to full supply
Dona_Ana	52,798,823	50,569,581	0.9578	50,569,791	0.9578	49,498,897	0.9375
Sierra	4981,082	4818,440	0.9673	4907,662	0.9853	4669,765	0.9375
Socorro	5689,667	5475,484	0.9624	5411,392	0.9511	5334,063	0.9375
Valencia	4267,322	4102,723	0.9614	4045,741	0.9481	4000,614	0.9375
Bernalillo	307,000	293,982	0.9576	293,660	0.9565	287,812	0.9375
Sandoval	702,092	676,010	0.9629	671,535	0.9565	658,211	0.9375
TOTAL	68,745,986	65,936,220	0.9591	65,899,781	0.9586	64,449,362	0.9375
Drought Loss	0	2809,766		2846,205		4296,624	

Table 8
Optimized Economic Value of Additional Water by New Mexico Rio Grande Basin County, Crop, Water Supply Scenario, and Water Trading Institution, 2022 ($/acre-ft).

County	Crop	Water Supply Scenario			
		full	drought		
			Water Trading Scope		
			unlimited	limited	none
Dona_Ana	1_pecan	0.00	68.67	68.66	96.22
Dona_Ana	2_alfalfa	0.00	68.67	68.66	87.17
Dona_Ana	3_corn	0.00	68.67	68.66	70.18
Dona_Ana	4_wheat	0.00	68.67	68.66	3.53
Dona_Ana	5_peppers	0.00	68.67	68.66	460.76
Dona_Ana	6_cotton	0.00	68.67	68.66	27.01
Dona_Ana	7_onions	0.00	68.67	68.66	399.74
Sierra	1_pecan	0.00	68.67	39.92	104.81
Sierra	2_alfalfa	0.00	68.67	39.92	89.74
Sierra	3_corn	0.00	68.67	39.92	70.62
Sierra	4_wheat	0.00	68.67	39.92	4.58
Sierra	5_peppers	0.00	68.67	39.92	468.00
Sierra	6_cotton	0.00	68.67	39.92	15.11
Sierra	7_onions	0.00	68.67	39.92	398.75
Socorro	1_pecan	0.00	68.67	79.20	107.18
Socorro	2_alfalfa	0.00	68.67	79.20	86.22
Socorro	3_corn	0.00	68.67	79.20	77.92
Socorro	4_wheat	0.00	68.67	79.20	1.73
Socorro	5_peppers	0.00	68.67	79.20	459.26
Socorro	6_cotton	0.00	68.67	79.20	16.79
Socorro	7_onions	0.00	68.67	79.20	404.82
Valencia	1_pecan	0.00	68.67	80.07	71.21
Valencia	2_alfalfa	0.00	68.67	80.07	87.90
Valencia	3_corn	0.00	68.67	80.07	69.54
Valencia	4_wheat	0.00	68.67	80.07	4.15
Valencia	5_peppers	0.00	68.67	80.07	459.84
Valencia	6_cotton	0.00	68.67	80.07	12.94
Valencia	7_onions	0.00	68.67	80.07	402.45
Bernalillo	1_pecan	0.00	68.67	69.59	88.36
Bernalillo	2_alfalfa	0.00	68.67	69.59	85.76
Bernalillo	3_corn	0.00	68.67	69.59	67.47
Bernalillo	4_wheat	0.00	68.67	69.59	4.64
Bernalillo	5_peppers	0.00	68.67	69.59	474.07
Bernalillo	6_cotton	0.00	68.67	69.59	17.31
Bernalillo	7_onions	0.00	68.67	69.59	402.90
Sandoval	1_pecan	0.00	68.67	74.60	83.98
Sandoval	2_alfalfa	0.00	68.67	74.60	88.73
Sandoval	3_corn	0.00	68.67	74.60	66.96
Sandoval	4_wheat	0.00	68.67	74.60	2.09
Sandoval	5_peppers	0.00	68.67	74.60	482.91
Sandoval	6_cotton	0.00	68.67	74.60	16.44
Sandoval	7_onions	0.00	68.67	74.60	398.93

outcomes.

Implementing a water trading policy across geographic areas such as counties and across agronomic scales like crops can allocate water more efficiently while controlling farm income losses. Startup costs face any effort to implement a water market, but after these startup costs to implement the program are present, the ongoing costs of daily implementation can be moderated (Griffin, 2016).

Findings from our work inform plans by farmers and policymakers who wish to allocate water efficiently in the sense of reducing the economic costs of water shortages. The results and approach taken in this model may have some adaptability to the case of transboundary water allocation conflicts. Our work also can inform water trading policy as an approach to enhance water economic efficiency and adaptability to face climate change or variability. More generally, the results from this work could also be a useful policy tool to control farm income losses where there are looming water shortages.

4.2. Scope and limitations

While this work examined the economic performance of various potential water trading institutions for handling shortages, it did not directly address water justice challenges connected to the various water trading arrangements (Franco et al., 2013; Zwarteveen and Boelens, 2014; Martinez-Alier et al., 2016; Sultana, 2018; Enqvist and Ziervogel, 2019), despite the importance of justice when considering impacts of water re-allocations facing the farming communities in our counties studied.

Water justice presents a number of difficulties in definition and implementation. Still, water trading has the potential to minimize the overall economic cost of water shortages, but that trading does not necessarily allocate the water to people who have the greatest need. A certain number of crop water uses might be assigned high priority for dietary reasons, such as security of access to food grains, a significant challenge in the arid and semi-arid regions of the Global South.

This model has a limited geographic and temporal scale. It is limited to one year, six crop irrigating counties and two irrigation districts in the Rio Grande watershed of North America. However, the approach presented in this work could see application to other irrigation regions of the western U.S. or to other countries internationally. We see a potential by which our approach could be adapted to observe other implementations or constraints associated with water trading. Examples include prioritizing shortage sharing to certain culturally important crops or prioritizing one irrigation district or county over another for cultural or historical reasons.

This work does not present a comprehensive plan of action on establishing and carrying out a water trading arrangement. Moreover, this work accounts for total water use based on measured ET without focusing whether used or traded water would come from a surface source or aquifer. The question of how to find more reliable meteorological data for assessing drought forecast to support water trading is not directly addressed here. Our work also does not distinguish between a water sale, water rental, water leasing, or a water right use option, all of which are important for understanding and predicting long versus short term changes in water use patterns for adapting to water shortages.

The nature of how water rights are defined and implemented also presents challenges. Completely defined property rights present an important element to carry out water trading wherever that trading occurs. A successful water trading arrangement needs professional water administration, or a proposed trading plan may confront economic, cultural, institutional, and even legal challenges.

Another important issue beyond the scope of this work is the issue of transactions costs. Some research has shown transactions costs can represent an important part of total costs of implementing any new policy, such as water trading. Since the model only provides information on the economic value of unlimited trading, a follow up question that emerges is whether the policy that provides the highest gains from trading (unlimited trading) is also the policy with lowest policy implementation costs.

This need not always be the case since unlimited trading may lead to significant third-party effects in regions of the world with a weak or non-existent water rights administrative system. Friction that increases transaction costs, both private (enforcing, policing) and institutional (lobbying, gathering of relevant data). Assessing these transactions costs is an important area for future work to better understand what policy is the most economically efficient.

4.3. Further work needed

Future research needs attention given to water justice, such as constraining water trades to assign certain minimum amounts of water to selected irrigators, such as specified crops like as food grains to support staple foods. Basic staples such as potatoes, pasta, rice, cereals, and bread are all examples of starchy foods and according to some medical

research need to constitute just over a third of the calories of a daily diet. Starchy (farinaceous) foods are dishes from pasta, noodles, and rice. These foods are highly nutritious and provide energy and dietary fiber, and include starchy flours, cereals, pulses, and starchy vegetables. So, water trading justice may assign an acceptable portion of acreage to these basic staples, such as wheat and corn for our region.

Our land/water allocation model is based on the principle of Ricardian rent, by which yields are highest on the best lands, but as the scale of production increases, yields fall off as the land quality is reduced. More work is needed to account for other water shortage adaptation methods such as water saving irrigation technologies as well a simple deficit irrigation, neither of which is addressed in the current work.

More future work is needed to distinguish water trading according to the hydrologic source from which the trade comes, either groundwater or surface water or both. In our study area, both surface and groundwater sources are used to supply irrigation water. Following the methods of USDA's Crop Layers database (U.S. Department of Agriculture, 2022b) as well as the OPEN ET water use tracking data source (OpenET, 2023), the model does not distinguish between these two sources. Additional future work is needed to bring attention to this limitation. Despite these limitations, water rights trading in New Mexico is routinely practiced, by which the right to pump new water from an aquifer or river by the buyer is matched by reduced pumping from that aquifer or river by the seller. Our method enables water trading by which the aquifer or surface water source or both is kept in hydrologic balance with and without a trading mechanism to adapt to water shortages. In effect, our model assumes frictionless reallocation of water when motivated by market forces. Physical and institutional limitations on these reallocations would reduce the capacity of water markets to address shortages. Future research could profitably establish a scenario in which water trading is restricted to the same water source, similar to our limited scenario in which water cannot be traded across county boundaries.

Among the investigated crops, pecan and alfalfa are permanent crops in terms of cultivation and harvest period. Still, growers have implemented water trading with both these crops. Pecans especially are almost always a water buyer than a seller because of its high value, so in drought periods, pecan acreages usually fall very little since these growers buy water from wherever they can find it to keep their acreages in production. Alfalfa growers often simply reduce the number of applications during a dry spell rather than taking much land out of production. Still because of the difficulty of reducing acreage for both pecans and alfalfa, some of the results of Table 4 could be unreliable.

Additionally, this kind of model could profitably be adapted to a range of dry regions internationally, as our model has considerable flexibility. More work on methods to guard against market failures of water trading arrangements is needed, partly because assessing the extent of the viability of implementing water trading policies is important. If it could be done, incorporating property rights into our model is a needed step forward to learn more about success in water trading institution that could be achieved.

5. Conclusions

Irrigated agriculture throughout much of the western U.S. relies on water resources that are under increasing stress from more frequent and severe droughts, reductions in snowpack, and alterations in the timing of runoff into surface water delivery systems. Much of the water used for irrigation in the western U.S. passes through irrigation organizations, which play an important role in the use and conservation of water resources through the maintenance of conveyance infrastructure, administration of water rights, and water allocation. Little work has been published to date systematically describing economic effects from alternative irrigation water trading arrangements for adapting to drought and climate change. We used USDA data on irrigation water use patterns in the central part of the Rio Grande Basin of the American Southwest to support formulation of a mathematical model predicting water use patterns to assess impacts on farm income associated with two water shortages and three water trading arrangements for handling these shortages. Model results show that unrestricted water trading minimizes the farm income lost from drought, while limited geographic scope trading also moderates farm income losses. Among shortage sharing methods, proportional sharing of shortages without trading produces the greatest loss in farm income for adapting to drought. These data suggest that water trading arrangements have an important role play in ensuring that there are available water supplies for irrigation when shortages occur. Our results provide science-informed guidance to water managers, stakeholders, and policymakers who need to discover irrigation management plans for arid and semi-arid regions internationally.

Disclaimer

The findings and conclusions in this publication are those of the author(s) and should not be construed to represent any official USDA or U.S. Government determination or policy.

Declaration of Competing Interest

The authors declare no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2024.109279.

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